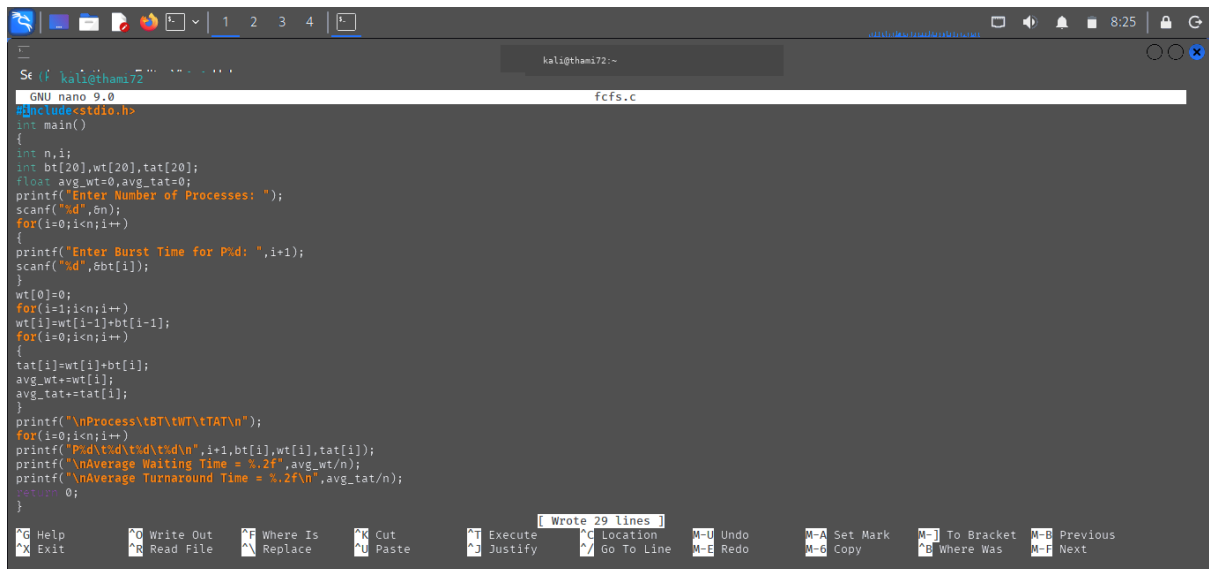
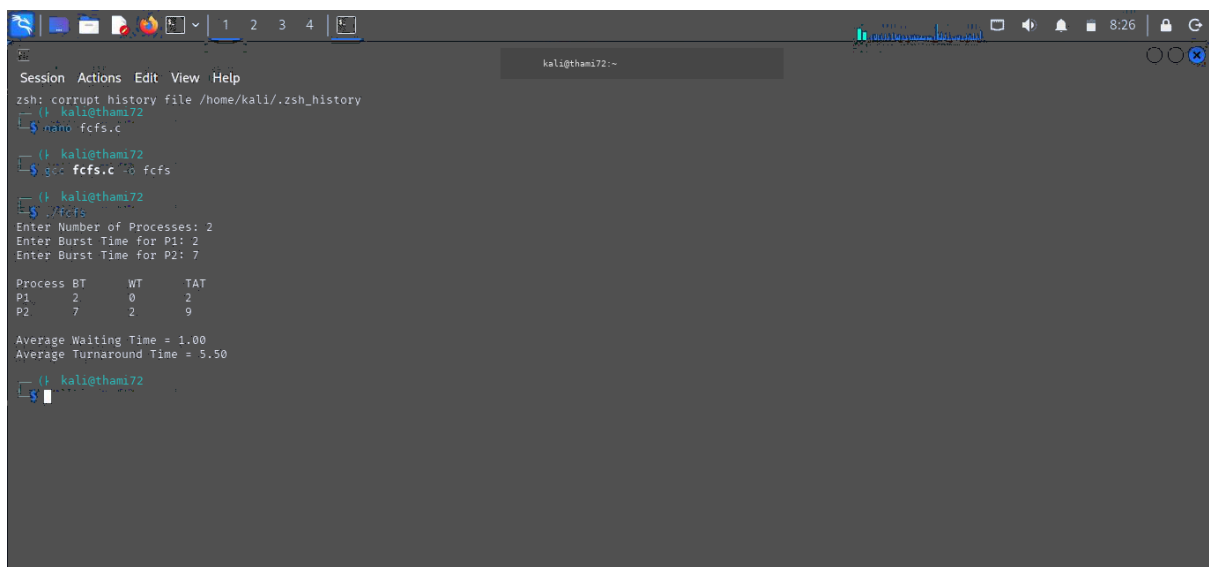


C PROGRAMMING : FCFS



```
GNU nano 9.0 fcfs.c
#include<stdio.h>
int main()
{
    int n,i;
    int bt[20],wt[20],tat[20];
    float avg_wt=0,avg_tat=0;
    printf("Enter Number of Processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("Enter Burst Time for P%d: ",i+1);
        scanf("%d",&bt[i]);
    }
    wt[0]=0;
    for(i=1;i<n;i++)
    {
        wt[i]=wt[i-1]+bt[i-1];
    }
    for(i=0;i<n;i++)
    {
        tat[i]=wt[i]+bt[i];
        avg_wt+=wt[i];
        avg_tat+=tat[i];
    }
    printf("\nProcess\tBT\tWT\tTAT\n");
    for(i=0;i<n;i++)
    {
        printf("P%d\t%d\t%d\t%d\n",i+1,bt[i],wt[i],tat[i]);
    }
    printf("\nAverage Waiting Time = %.2f",avg_wt/n);
    printf("\nAverage Turnaround Time = %.2f\n",avg_tat/n);
    return 0;
}
```

OUTPUT :

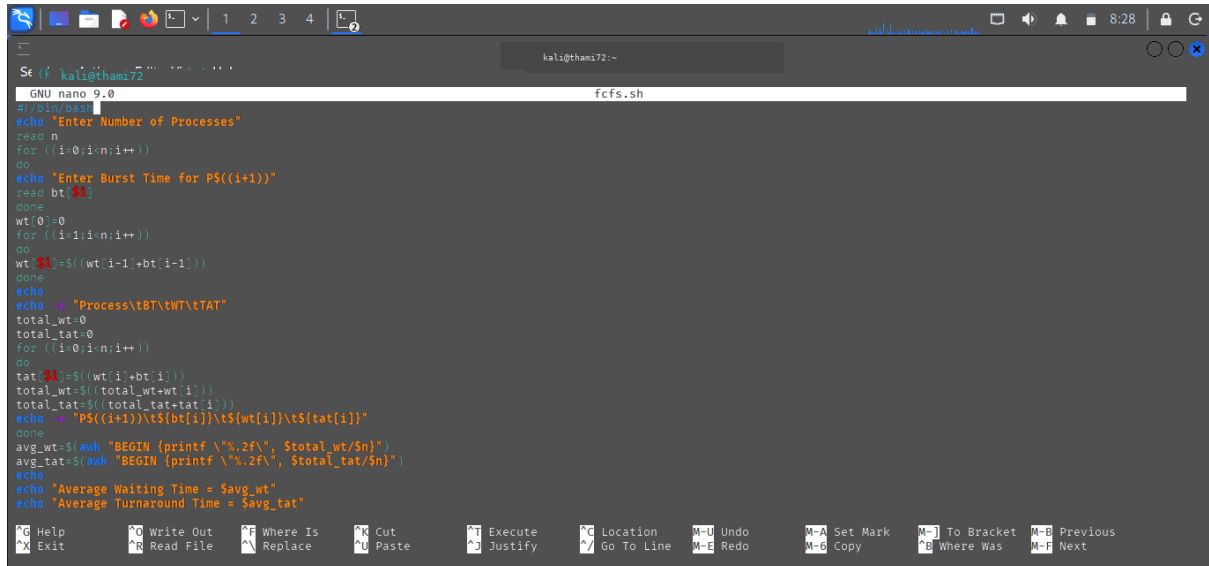


```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72:~$ nano fcfs.c
kali@thami72:~$ gcc fcfs.c -o fcfs
kali@thami72:~$ ./fcfs
Enter Number of Processes: 2
Enter Burst Time for P1: 2
Enter Burst Time for P2: 7

Process BT    WT    TAT
P1      2      0      2
P2      7      2      9

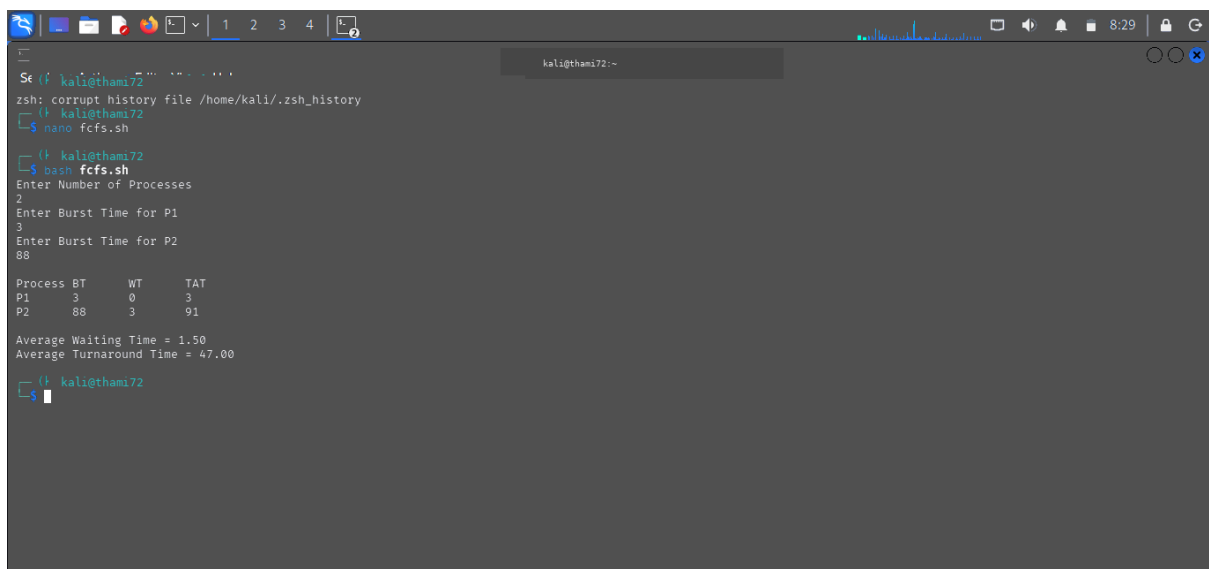
Average Waiting Time = 1.00
Average Turnaround Time = 5.50
kali@thami72:~$
```

SHELL PROGRAMMING :FCFS



```
Se (f kali@thami72 ~)
GNU nano 9.0 fcfs.sh
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i=0;i<n;i++))
do
echo "Enter Burst Time for P$((i+1))"
read bt[i]
done
wt[0]=0
for ((i=1;i<n;i++))
do
wt[i]=$((wt[i-1]+bt[i-1]))
done
echo -e "Process\tBT\tWT\tTAT"
total_wt=0
total_tat=0
for ((i=0;i<n;i++))
do
tat[i]=$((wt[i]+bt[i]))
total_wt=$((total_wt+wt[i]))
total_tat=$((total_tat+tat[i]))
echo -e "P$((i+1))\t${bt[i]}\t${wt[i]}\t${tat[i]}"
done
avg_wt=$(awk "BEGIN {printf \"%.2f\", $total_wt/$n}")
avg_tat=$(awk "BEGIN {printf \"%.2f\", $total_tat/$n}")
echo
echo "Average Waiting Time = $avg_wt"
echo "Average Turnaround Time = $avg_tat"
```

OUTPUT :

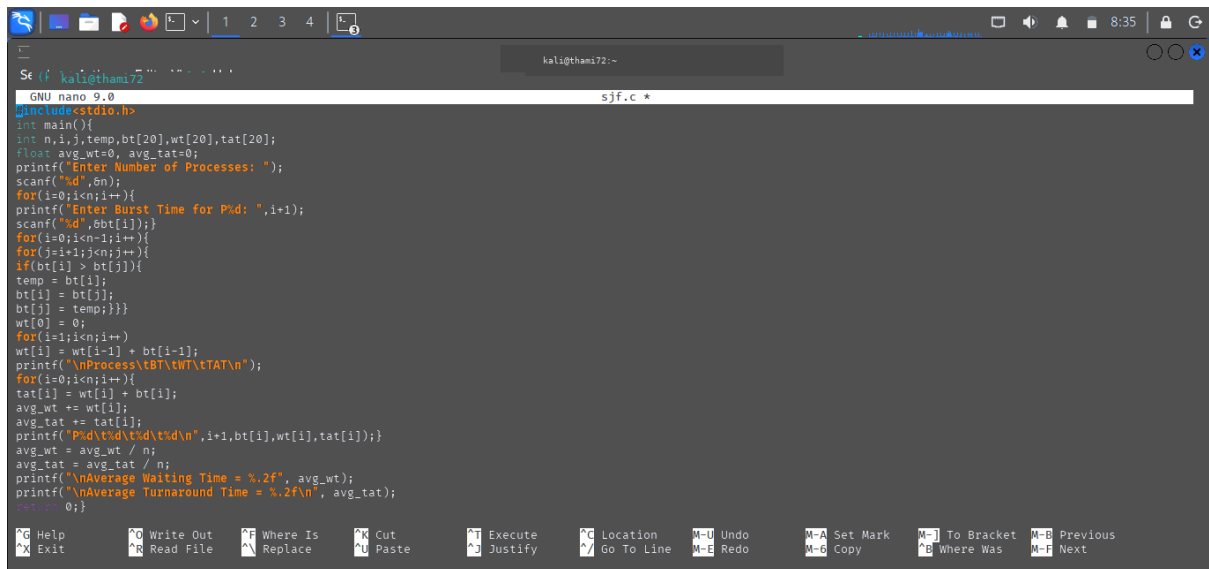


```
Se (f kali@thami72 ~)
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano fcfs.sh
kali@thami72 ~
$ bash fcfs.sh
Enter Number of Processes
2
Enter Burst Time for P1
3
Enter Burst Time for P2
88

Process BT      WT      TAT
P1       3       0       3
P2      88       3      91

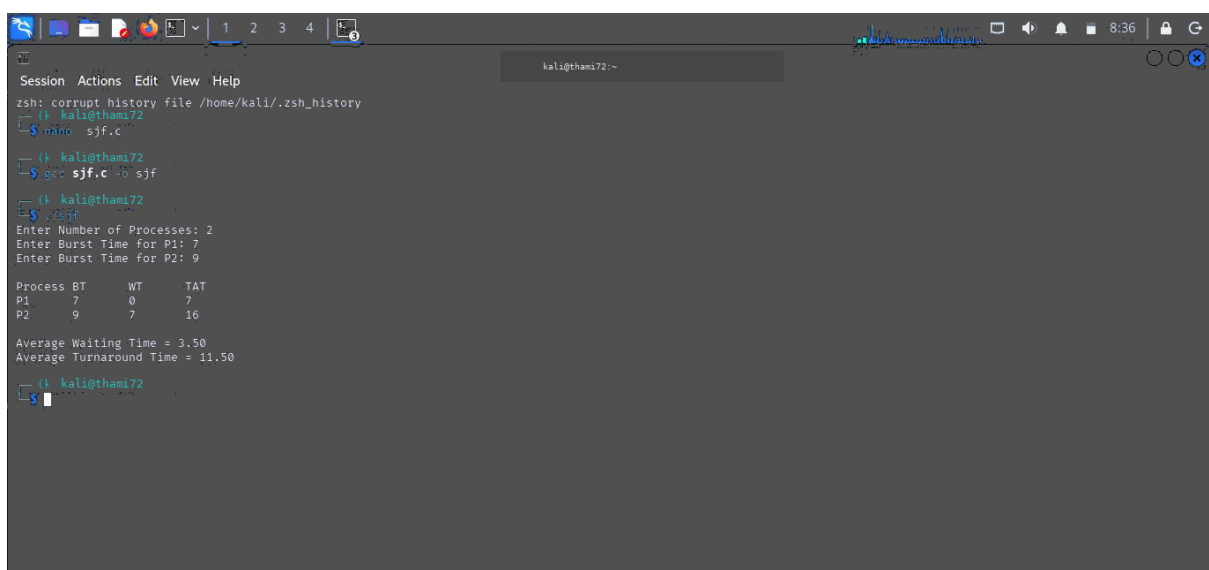
Average Waiting Time = 1.50
Average Turnaround Time = 47.00
kali@thami72 ~
$
```

C PROGRAMMING :SJF



```
GNU nano 9.0 sjf.c *
#include<stdio.h>
int main(){
    int n,i,j,temp,bt[20],wt[20],tat[20];
    float avg_wt=0, avg_tat=0;
    printf("Enter Number of Processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        printf("Enter Burst Time for P%d: ",i+1);
        scanf("%d",&bt[i]);
        for(j=i+1;j<n;j++){
            if(bt[i] > bt[j]){
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
        wt[0] = 0;
        for(i=1;i<n;i++){
            wt[i] = wt[i-1] + bt[i-1];
            printf("\nProcess\tBT\tWT\tTAT\n");
            for(i=0;i<n;i++){
                tat[i] = wt[i] + bt[i];
                avg_wt += wt[i];
                avg_tat += tat[i];
                printf("P%d\t%d\t%d\t%d\n",i+1,bt[i],wt[i],tat[i]);
            }
            avg_wt = avg_wt / n;
            avg_tat = avg_tat / n;
            printf("\nAverage Waiting Time = %.2f", avg_wt);
            printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
            return 0;
        }
    }
```

OUTPUT :

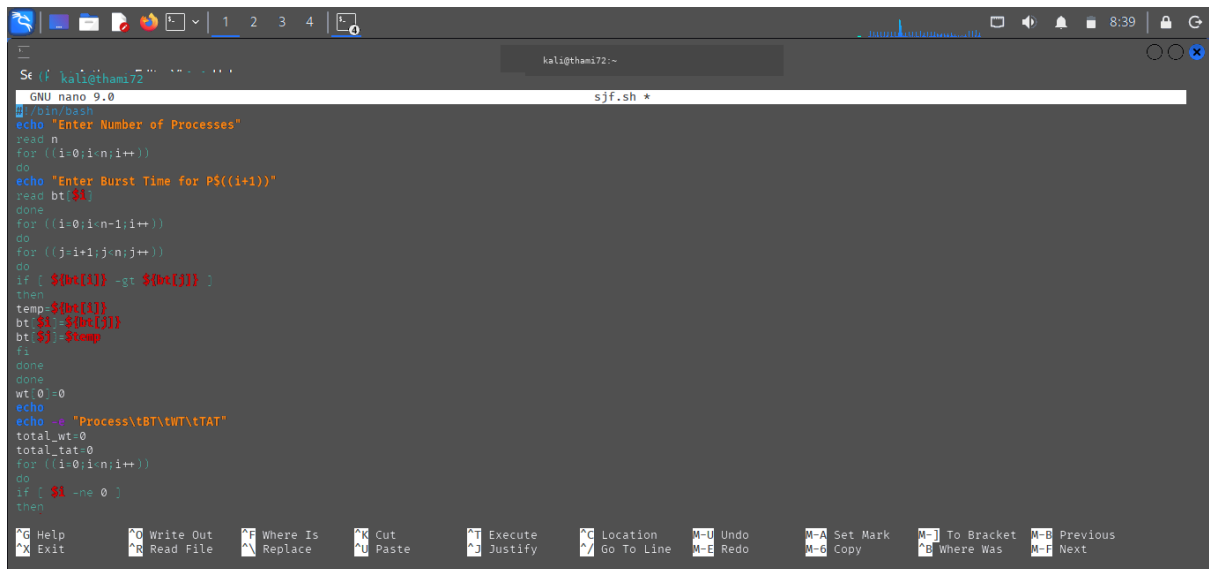


```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72:~$ nano sjf.c
kali@thami72:~$ gcc sjf.c -o sjf
kali@thami72:~$ ./sjf
Enter Number of Processes: 2
Enter Burst Time for P1: 7
Enter Burst Time for P2: 9

Process BT    WT    TAT
P1      7      0      7
P2      9      7     16

Average Waiting Time = 3.50
Average Turnaround Time = 11.50
kali@thami72:~$
```

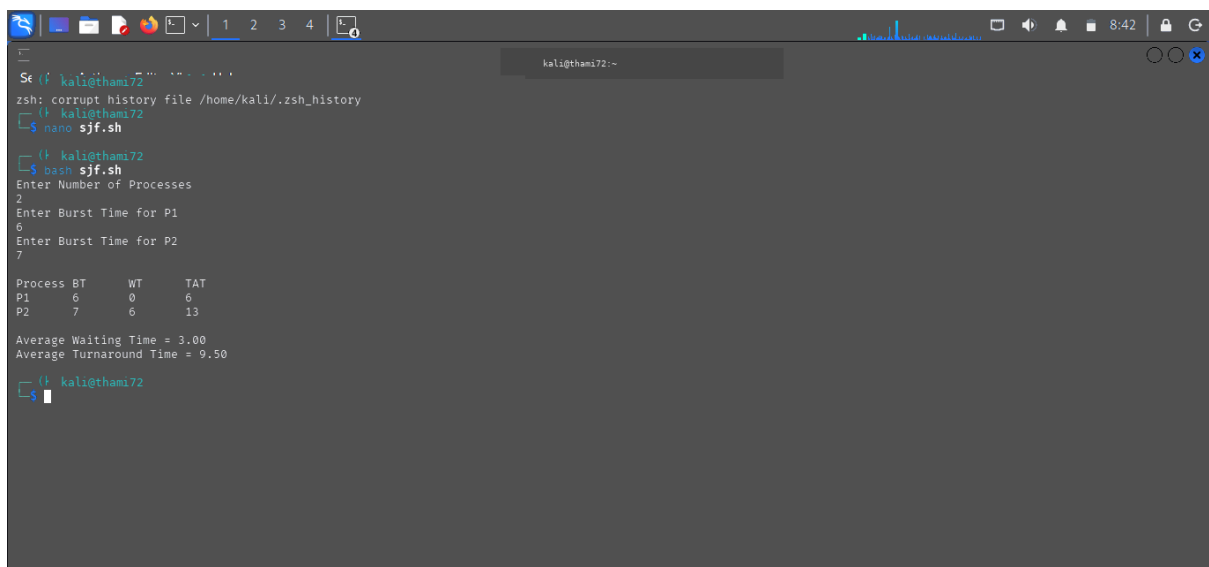
SHELL PROGRAMMING : SJF



```
GNU nano 9.0 sjf.sh *
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i=0;i<n;i++))
do
echo "Enter Burst Time for P$((i+1))"
read bt[$i]
done
for ((i=0;i<n-1;i++))
do
for ((j=i+1;j<n;j++))
do
if [ ${bt[$i]} -gt ${bt[$j]} ]
then
temp=${bt[$i]}
bt[$i]=${bt[$j]}
bt[$j]=$temp
fi
done
done
wt=0
echo -e "Process\tBT\tWT\tTAT"
total_wt=0
total_tat=0
for ((i=0;i<n;i++))
do
if [ ${bt[$i]} -ne 0 ]
then

```

OUTPUT :



```
Se (f kali@thami72 ~)
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72 ~
$ nano sjf.sh
kali@thami72 ~
$ bash sjf.sh
Enter Number of Processes
2
Enter Burst Time for P1
6
Enter Burst Time for P2
7
Process BT    WT    TAT
P1      6      0      6
P2      7      6     13
Average Waiting Time = 3.00
Average Turnaround Time = 9.50
kali@thami72 ~
$
```

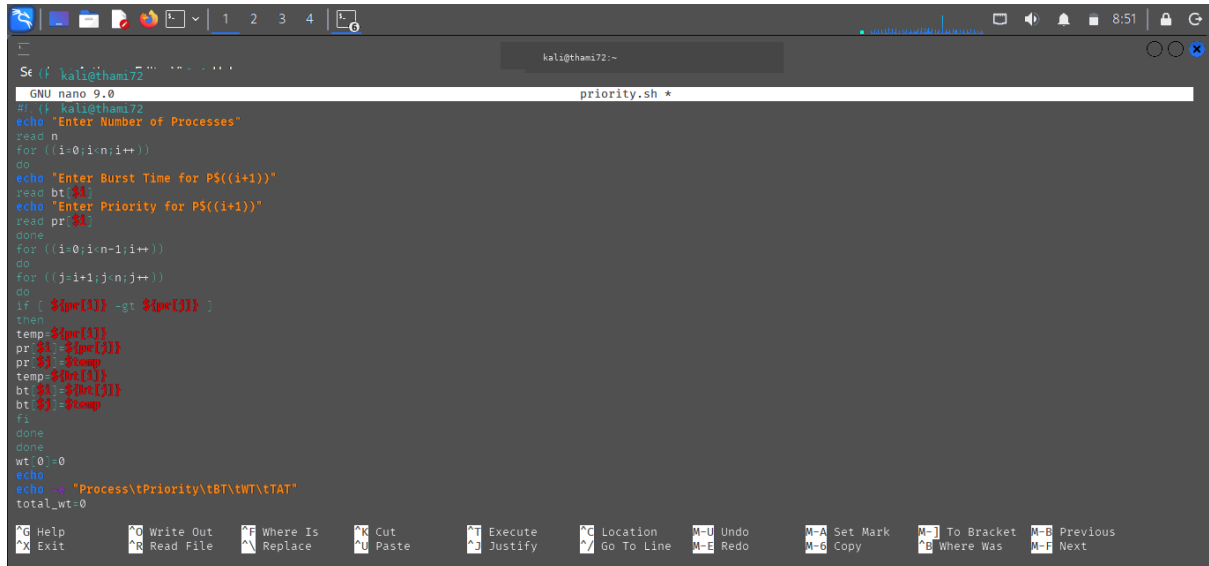
C PROGRAMMING : PRIORITY SCHEDULING

```
GNU nano 9.0 priority.c *
#include<stdio.h>
int main() {
    int n,i,j,temp, bt[20],pr[20],wt[20],tat[20];
    float avg_wt=0, avg_tat=0;
    printf("Enter Number of Processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        printf("\nEnter Burst Time for P%d: ",i+1);scanf("%d",&bt[i]);
        printf("Enter Priority for P%d: ",i+1);scanf("%d",&pr[i]);
        for(i=0;i<n-1;i++){
            for(j=i+1;j<n;j++){
                if(pr[i] > pr[j]){
                    temp = pr[i];
                    pr[i] = pr[j];
                    pr[j] = temp;
                    temp = bt[i];
                    bt[i] = bt[j];
                    bt[j] = temp;}}wt[0] = 0;
                for(i=1;i<n;i++){
                    wt[i] = wt[i-1] + bt[i-1];
                    printf("\nProcess\tPriority\tBT\tWT\tTAT\n");
                    for(i=0;i<n;i++){
                        tat[i] = wt[i] + bt[i];
                        avg_wt += wt[i];avg_tat += tat[i];
                        printf("P%d\t%d\t%d\t%d\t%d\n",i+1, pr[i], bt[i], wt[i], tat[i]);
                    }
                    avg_wt /= n;avg_tat /= n;
                    printf("\nAverage Waiting Time = %.2f", avg_wt);
                    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
                }
                return 0;
            }
```

OUTPUT :

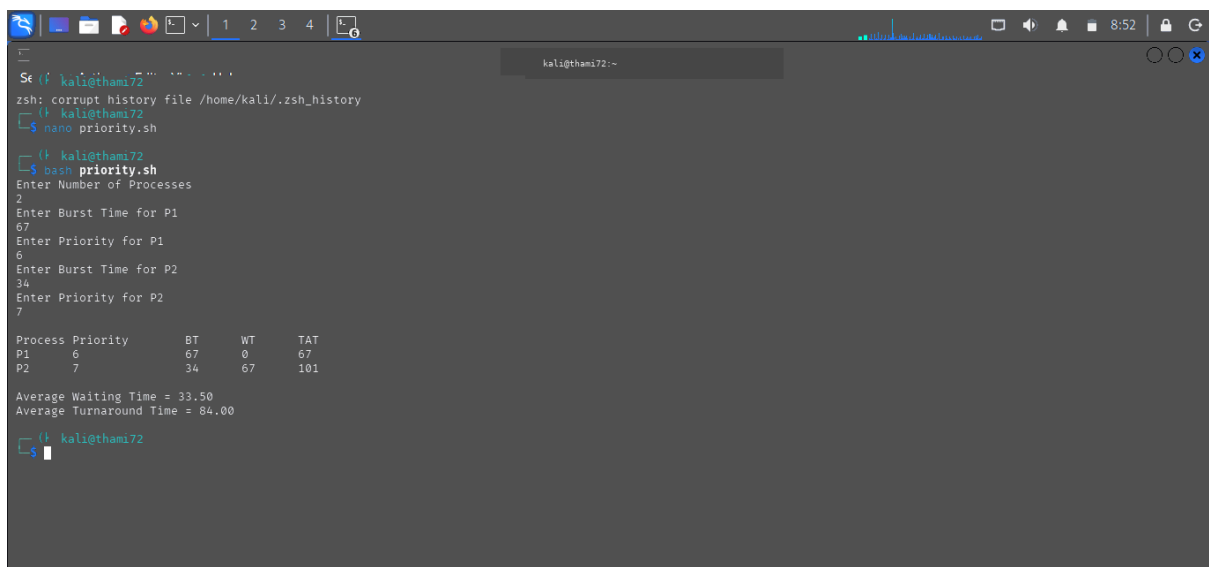
```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72:~$ nano priority.c
kali@thami72:~$ gcc priority.c -o priority
kali@thami72:~$ ./priority
Enter Number of Processes: 2
Enter Burst Time for P1: 55
Enter Priority for P1: 4
Enter Burst Time for P2: 76
Enter Priority for P2: 5
Process Priority      BT      WT      TAT
P1          4        55         0       55
P2          5        76        55      131
Average Waiting Time = 27.50
Average Turnaround Time = 93.00
kali@thami72:~$
```

SHELL PROGRAMMING : SJF



```
GNU nano 9.0 priority.sh
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i=0;i<n;i++))
do
echo "Enter Burst Time for P$((i+1))"
read bt[$i]
echo "Enter Priority for P$((i+1))"
read pr[$i]
done
for ((i=0;i<n-1;i++))
do
for ((j=i+1;j<n;j++))
do
if [ ${pr[$i]} -gt ${pr[$j]} ]
then
temp=${pr[$i]}
pr[$i]=${pr[$j]}
pr[$j]=$temp
temp=${bt[$i]}
bt[$i]=${bt[$j]}
bt[$j]=$temp
fi
done
done
wt=0
echo
echo -e "Process\tPriority\tBT\tWT\tTAT"
total_wt=0
```

OUTPUT :

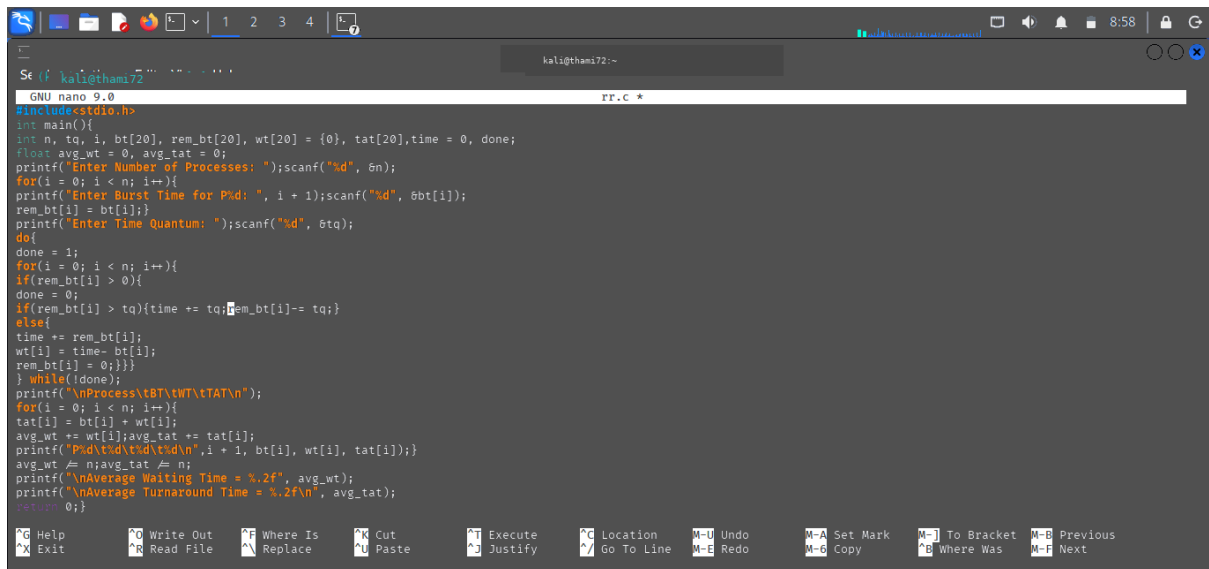


```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72:~$ nano priority.sh
kali@thami72:~$ bash priority.sh
Enter Number of Processes
2
Enter Burst Time for P1
67
Enter Priority for P1
6
Enter Burst Time for P2
34
Enter Priority for P2
7

Process Priority      BT      WT      TAT
P1         6       67       0       67
P2         7       34       67      101

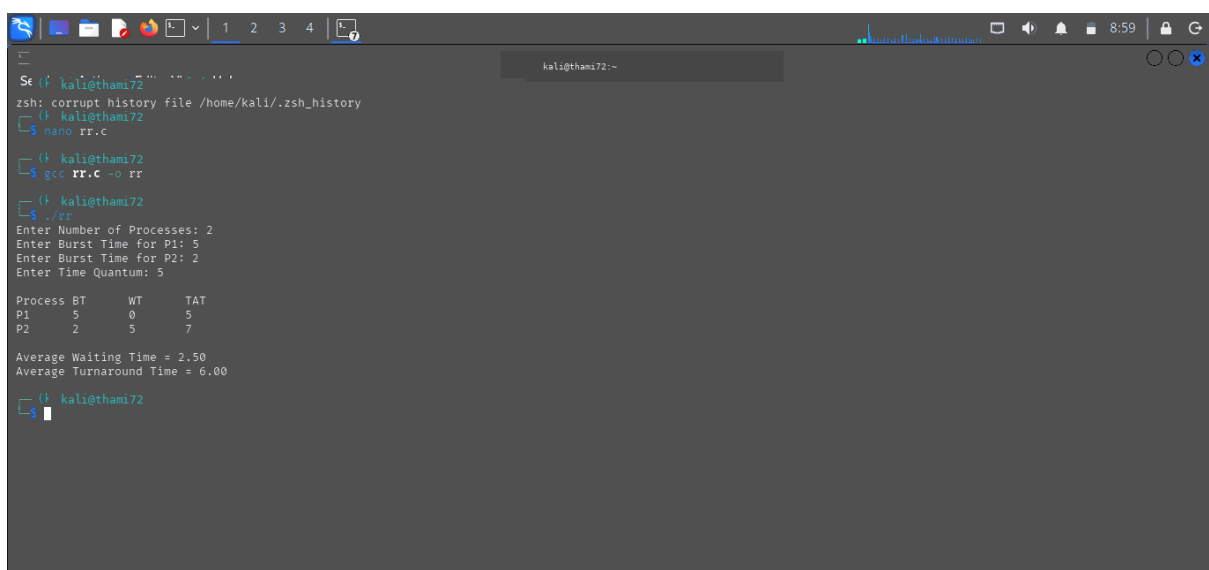
Average Waiting Time = 33.50
Average Turnaround Time = 84.00
kali@thami72:~$
```

C PROGRAMMING : ROUND ROBIN



```
GNU nano 9.0 rr.c *
#include<stdio.h>
int main(){
    int n, tq, i, bt[20], rem_bt[20], wt[20] = {0}, tat[20], time = 0, done;
    float avg_wt = 0, avg_tat = 0;
    printf("Enter Number of Processes: ");scanf("%d", &n);
    for(i = 0; i < n; i++){
        printf("Enter Burst Time for P%d: ", i + 1);scanf("%d", &bt[i]);
        rem_bt[i] = bt[i];
        printf("Enter Time Quantum: ");scanf("%d", &tq);
    }
    do{
        done = 1;
        for(i = 0; i < n; i++){
            if(rem_bt[i] > 0){
                done = 0;
                if(rem_bt[i] > tq){time += tq; rem_bt[i] -= tq;}
                else{
                    time += rem_bt[i];
                    wt[i] = time - bt[i];
                    rem_bt[i] = 0;
                }
            }
        }
        while(!done);
        printf("\nProcess\tBT\tWT\tTAT\n");
        for(i = 0; i < n; i++){
            tat[i] = bt[i] + wt[i];
            avg_wt += wt[i];avg_tat += tat[i];
            printf("P%d\t%d\t%d\t%d\n", i + 1, bt[i], wt[i], tat[i]);
        }
        avg_wt /= n;avg_tat /= n;
        printf("\nAverage Waiting Time = %.2f", avg_wt);
        printf("\nAverage Turnaround Time = %.2f\n", avg_tat);
    }
    return 0;
}
```

OUTPUT :

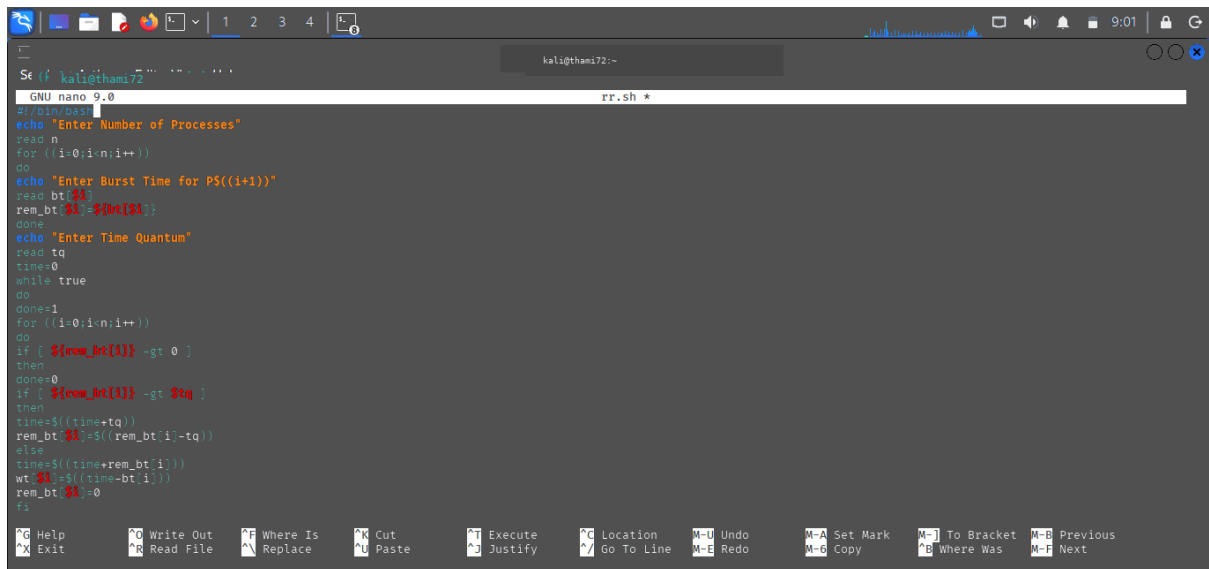


```
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72:~$ nano rr.c
kali@thami72:~$ gcc rr.c -o rr
kali@thami72:~$ ./rr
Enter Number of Processes: 2
Enter Burst Time for P1: 5
Enter Burst Time for P2: 2
Enter Time Quantum: 5

Process BT      WT      TAT
P1      5        0        5
P2      2        5        7

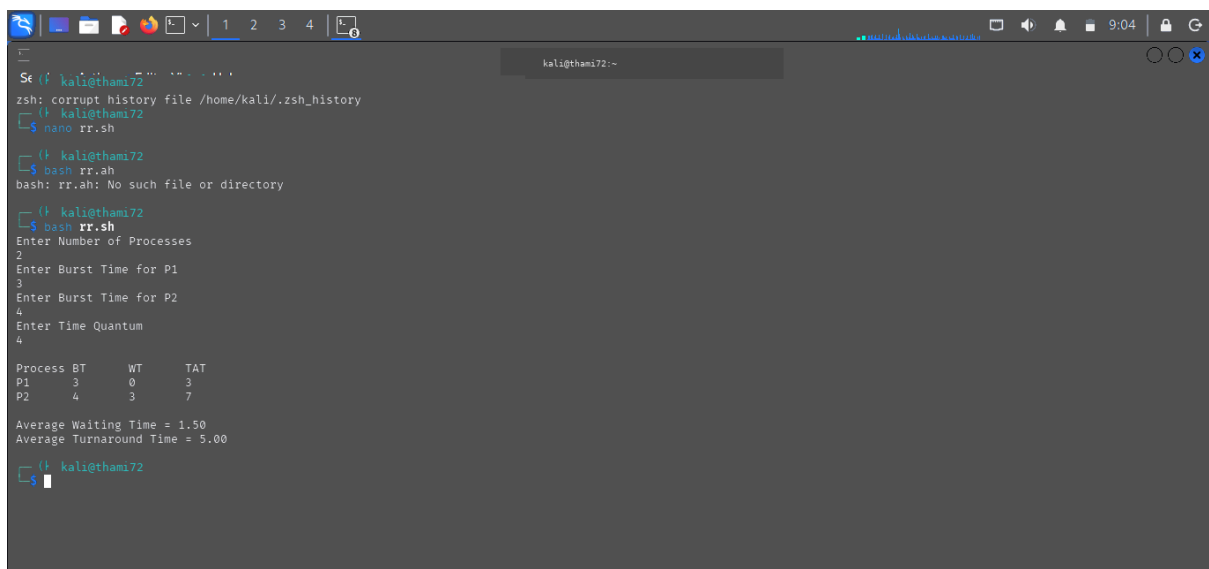
Average Waiting Time = 2.50
Average Turnaround Time = 6.00
kali@thami72:~$
```

SHELL PROGRAMMING : ROUND ROBIN



```
GNU nano 9.0 rr.sh *
#!/bin/bash
echo "Enter Number of Processes"
read n
for ((i=0;i<n;i++))
do
echo "Enter Burst Time for P$((i+1))"
read bt[$i]
rem_bt[$i]=$bt[$i]
done
echo "Enter Time Quantum"
read tq
time=0
while true
do
done=1
for ((i=0;i<n;i++))
do
if [ ${rem_bt[$i]} -gt 0 ]
then
done=0
if [ ${rem_bt[$i]} -gt $tq ]
then
time=$((time+tq))
rem_bt[$i]=$((rem_bt[$i]-tq))
else
time=$((time+rem_bt[$i]))
wt[$i]=$((time-bt[$i]))
rem_bt[$i]=0
fi
fi
done
done
done
```

OUTPUT :



```
kali@thami72:~$ nano rr.sh
zsh: corrupt history file /home/kali/.zsh_history
kali@thami72:~$ bash rr.sh
bash: rr.sh: No such file or directory
kali@thami72:~$ bash rr.sh
Enter Number of Processes
2
Enter Burst Time for P1
3
Enter Burst Time for P2
4
Enter Time Quantum
4

Process BT    WT    TAT
P1      3      0      3
P2      4      3      7

Average Waiting Time = 1.50
Average Turnaround Time = 5.00
kali@thami72:~$
```